

Parallax-Free X-ray Area Detector With High Spatial Resolution

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The new parallax-free X-ray 2D detector (SD-1000) with the pixel array of 1024 x 1024 has been designed and tested. This detector is the basic element of the X-ray diffractometer which is constructed for the investigation of biological macromolecules, viruses, liquid crystals and powders. The first test results are presented.

1. Introduction

More than 200 samples of protein single crystals have been investigated in the last few years with the X-ray area detectors based on various versions of multiwire chambers which have been constructed at the Laboratory of High Energies, JINR in collaboration with the Institute of Crystallography [1-3]. The space structure of several proteins has been reconstructed as follows:

- T-cataza, $M=200.000$, $a=106$ Å, resolution $\sim 2,7$ Å;
- CMT_v -virus, $M=7.800.000$, $a=480$ Å, resolution ~ 5.0 Å.

The new X-ray area detector with high spatial resolution has been developed by the same collaboration. The device consists of

- parallax-free X-ray area detector,
- readout system based on fast cathode delay lines,
- electronics of data acquisition with fast TDC and incremented memory.

The detector with a spherical Be-window and drift space (with radial field lines) has the following advantages [4] :

- Suppression of parallax for inclined beams;
- X-ray photons are converted to photoelectrons with 100% efficiency;
- randomization of radiation time structure due to the thickness of drift gap and low velocity of electrons;

- diffusion of the primary electron cloud along the drift space which provides the same spatial resolution on both cathode planes.

2. X-ray area detector

Schematic drawing of the detector is shown in fig. 1.

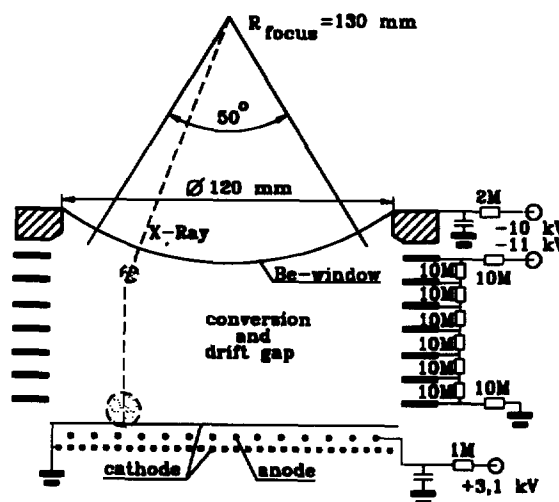


Figure 1. Schematic drawing of the parallax-free X-ray area detector with a spherical Be-window

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The MWPC with an anode-cathode distance 2 mm and an anode pitch 1 mm is placed at the sealed pressurized vessel with the beryllium entrance window 1 mm thick and 120 mm in diameter, shaped like a sphere with 120 mm radius. The metal rings (with a distributed electrical potential) are placed around the ~ 6 cm drift space. The detector volume is filled with the gas mixture of Xe(81%), Ar(8%), CH₄(11%) at a ~ 3 bar.

During the drift of electron cloud its size is enlarged and thus both spatial resolution and the channel uniformity across anode wires improve significantly.

Coordinate information is read out by means of strip delay lines with $T=250$ ns. The data acquisition system with the fast TDC (operating in real time) and the buffer incremented memory of $1024 \times 1024 \times 16$ bits is used for coordinates measurements. The arithmetic sum discriminator (based on the comparison of the sum of two time intervals between signals from different end of delay line and anode signal) permit to work with a maximum count rate of ~ 1 MHz.

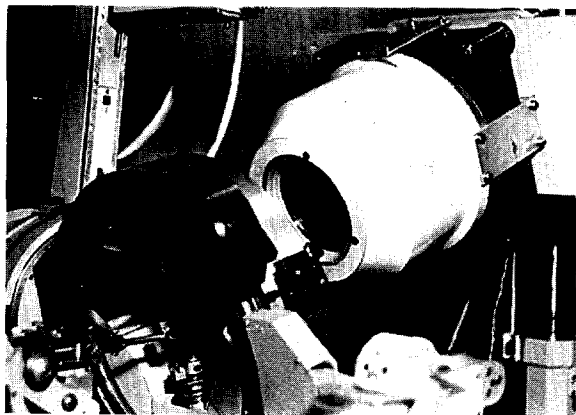


Figure 2. X-ray diffractometer CARD-7 with the area detector SD-1000

3. Basic detector characteristics

The detector was initially tested under laboratory conditions of LHE JINR and then at the Institute of Crystallography. The basic characteristics of detector are as follows:

- Detection efficiency - 80% (8 keV);
- Differential nonlinearity - 1% (Y), 3% (X);
- Intrinsic resolution - 170×170 μm ;
- Number of pixels - 1024×1024 or 512×512 ;
- Count rate (30% losses) - 3×10^5 1/s;¹
- Local count rate $\sim 10^4$ 1/s;
- Solid angle (without parallax) - 50° ;
- Noise - 30 1/s;
- Dynamic range - $> 10^6$;

This area detector is the base of the X-ray diffractometer CARD-7 (fig. 2) used for powder and protein investigations.



Figure 3. The diffraction pattern of the $Y_3Al_2(AlO_4)_3$ powder, obtained with X-ray diffractometer CARD-7

The diffraction pattern of the powder's sample is presented in fig. 3

¹A count rate with additional TDC $\sim 6 \times 10^5$ 1/s

4. Conclusions

The new high resolution parallax-free X-ray area detector with high uniformity of efficiency and low differential nonlinearity is constructed, tested and first structure investigations are started with an X-ray diffractometer. The detector operation is very stable.

5. Acknowledgements

The authors would like to thank Prof. A.M.Baldin for supporting this work and Yu.G.Fedulov, N.P.Volkov, A.E.Moskovsky, V.A.Beliakov for their help in the fabrication of the detector.

REFERENCES

1. Yu.V.Zanevsky, G.A.Cheremukhina, S.P.Chernenko et al., Nucl. Instr. and Meth. A310 (1991), 392
2. Yu.V.Zanevsky, S.P.Chernenko, D.M.Kheiker et al., Kristallografia 32(2) (1993), 252
3. S.P.Chernenko, G.A.Cheremukhina, O.V.Fateev et al., Nucl. Instr. and Meth. A348 (1994), 261
4. R.Fourme, Report of the International Workshop on Area Detectors, Harima Science Garden City, Hyogo, Japan November 24-26, 1993, 35